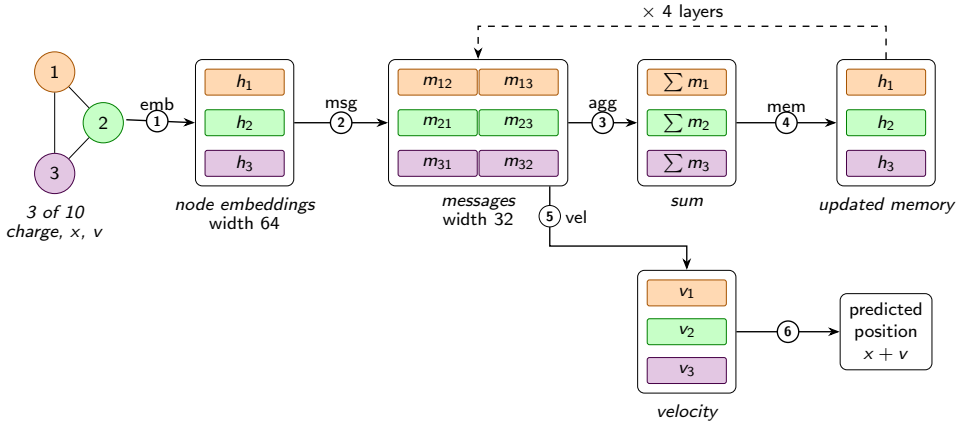
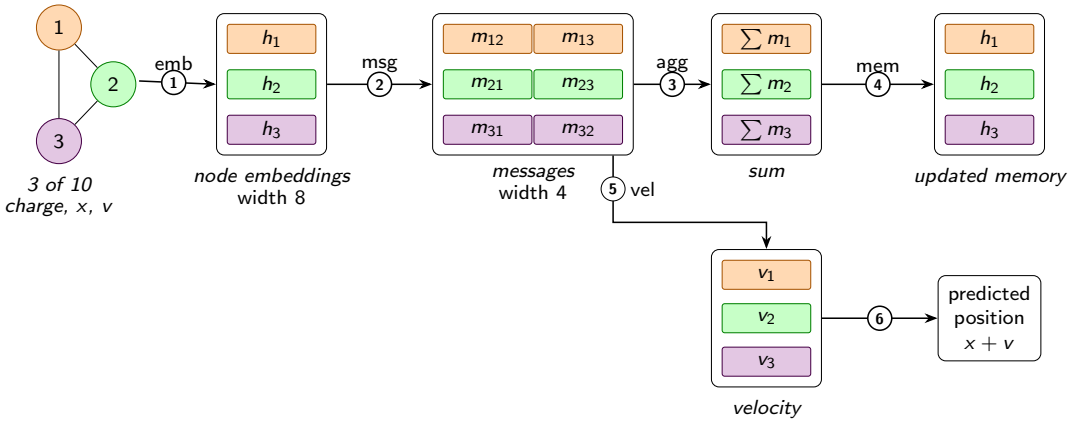
EGNN, big 4 layers, hidden 64, message 32



$$m_{ij} = \mathrm{ReLU}(Wh_i + Uh_j + V\|x_i - x_j\|^2)$$

$$v \leftarrow v + \textstyle\sum_j \tanh(w^\top m_{ij})\,(x_i - x_j) \qquad h \leftarrow h + \mathrm{ReLU}(W[h, \textstyle\sum_j m_{ij}])$$

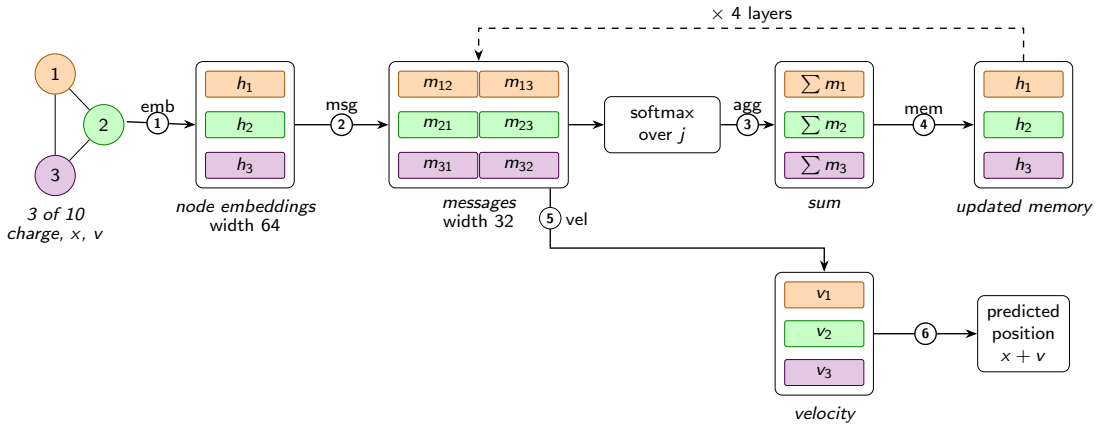
EGNN, small 1 layer, hidden 8, message 4



$$m_{ij} = \mathrm{ReLU}(Wh_i + Uh_j + V\|x_i - x_j\|^2)$$

$$v \leftarrow v + \sum_j \tanh(w^\top m_{ij})\,(x_i - x_j) \qquad h \leftarrow h + \mathrm{ReLU}(W[h, \textstyle\sum_j m_{ij}])$$

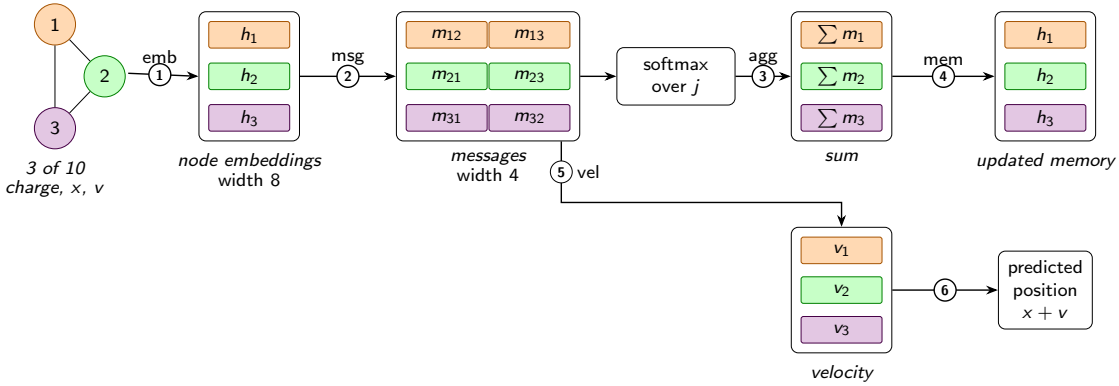
Attention EGNN, big 4 layers, hidden 64, message 32



$$m_{ij} = \mathrm{ReLU}(Wh_i + Uh_j + V\|x_i - x_j\|^2)$$

$$v \leftarrow v + \textstyle\sum_j \tanh(w^\top m_{ij})\,(x_i - x_j) \qquad h \leftarrow h + \mathrm{ReLU}(W[h, \textstyle\sum_j m_{ij}])$$

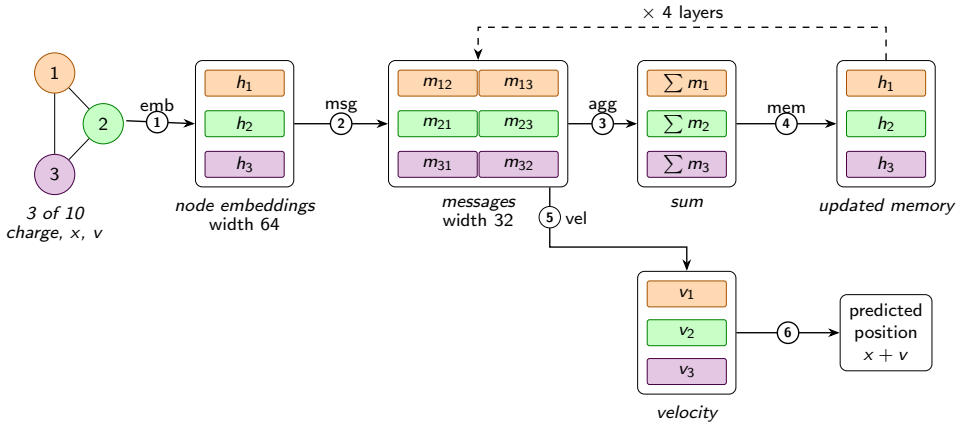
Attention EGNN, small 1 layer, hidden 8, message 4



$$m_{ij} = \mathrm{ReLU}(Wh_i + Uh_j + V\|x_i - x_j\|^2)$$

$$v \leftarrow v + \textstyle\sum_j \tanh(w^\top m_{ij})\,(x_i - x_j) \qquad h \leftarrow h + \mathrm{ReLU}(W[h, \textstyle\sum_j m_{ij}])$$

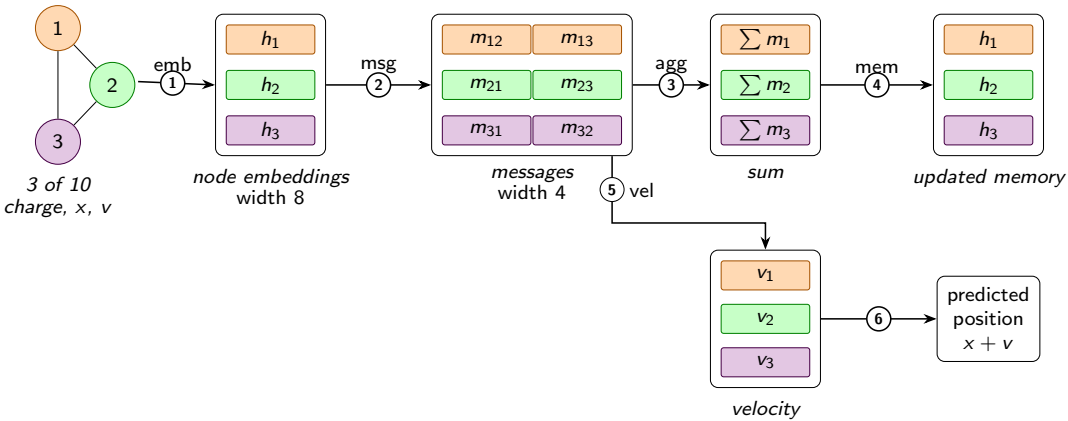
Non-equivariant GNN, big 4 layers, hidden 64, message 32



$$m_{ij} = \mathrm{ReLU}(Wh_i + Uh_j + V\|x_i - x_j\|^2 + A[x_i, x_j])$$

$$v \leftarrow v + \textstyle\sum_j \tanh(w^\top m_{ij})\,(x_i - x_j) \qquad h \leftarrow h + \mathrm{ReLU}(W[h, \textstyle\sum_j m_{ij}])$$

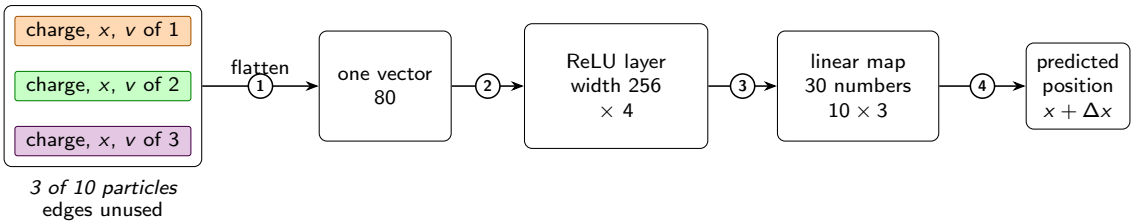
Non-equivariant GNN, small 1 layer, hidden 8, message 4



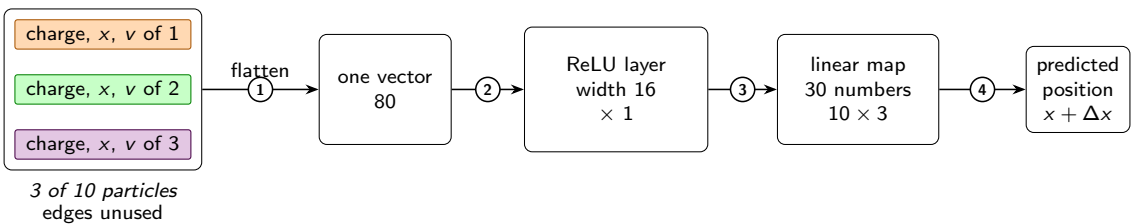
$$m_{ij} = \mathrm{ReLU}(Wh_i + Uh_j + V\|x_i - x_j\|^2 + A[x_i, x_j])$$

$$v \leftarrow v + \textstyle\sum_j \tanh(w^\top m_{ij})\,(x_i - x_j) \qquad h \leftarrow h + \mathrm{ReLU}(W[h, \textstyle\sum_j m_{ij}])$$

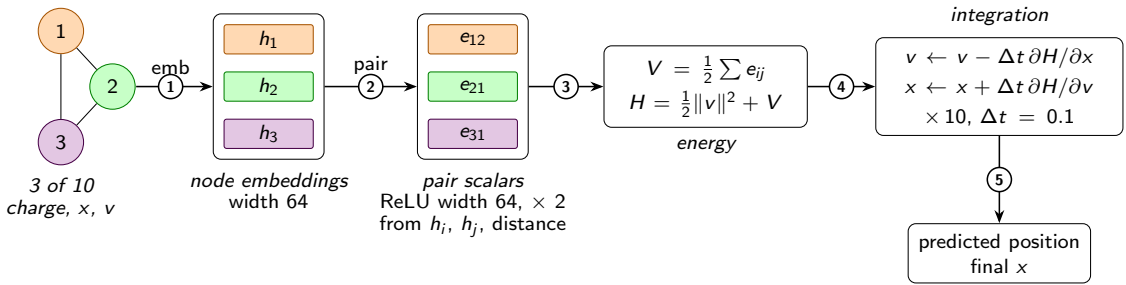
MLP, big 4 layers, hidden 256



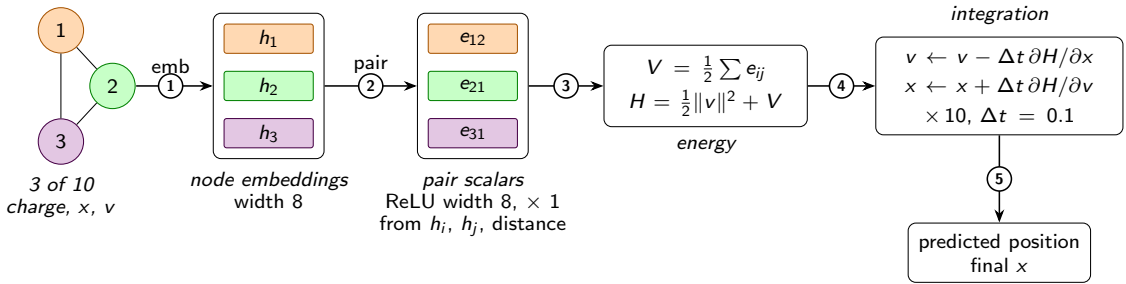
MLP, small 1 layer, hidden 16



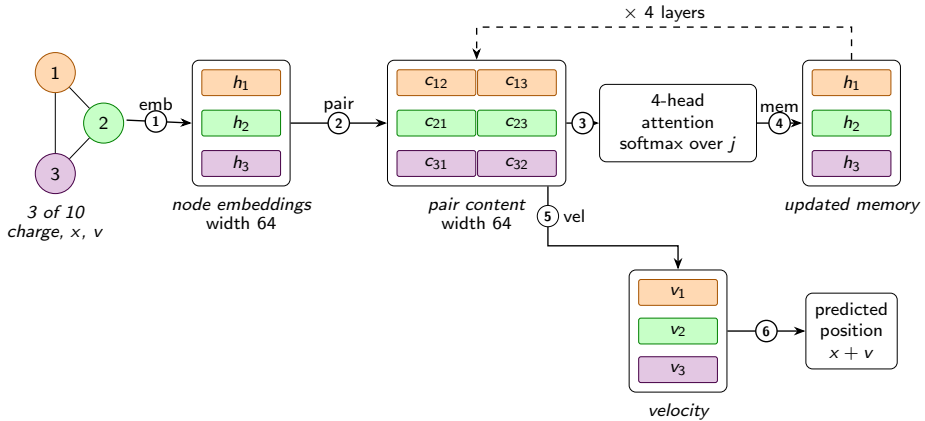
Hamiltonian network, big potential width 64, 2 layers, 10 steps



Hamiltonian network, small potential width 8, 1 layer, 10 steps



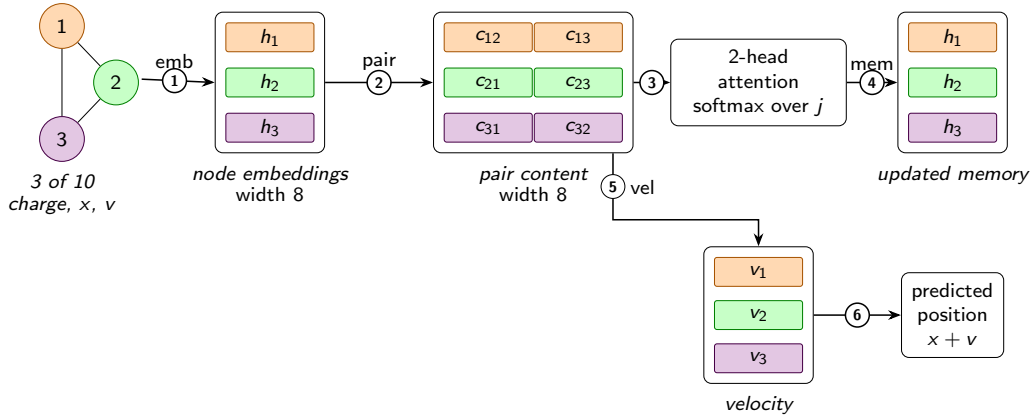
Equivariant transformer, big 4 layers, hidden 64, 4 heads



$$c_{ij} = Wh_i + Uh_j + V\|x_i - x_j\|^2 \qquad h \leftarrow h + \mathrm{ReLU}(W[h, \textstyle\sum_j \bar{\alpha}_{ij} c_{ij}])$$

$$v \leftarrow v + \textstyle\sum_j \tanh(u^\top c_{ij})\, \bar{\alpha}_{ij}\, (x_i - x_j) \qquad \bar{\alpha}_{ij}\text{: attention weight, averaged over heads}$$

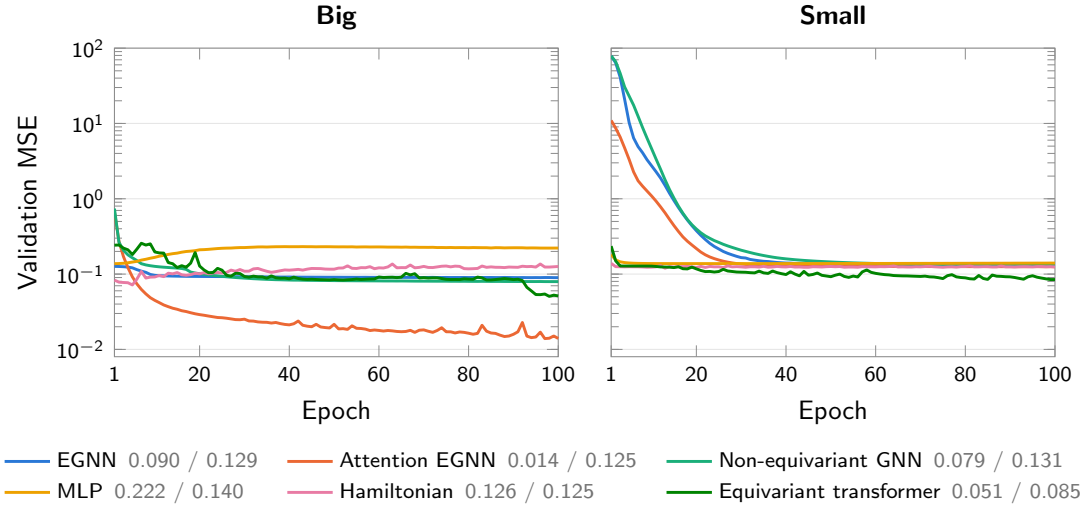
Equivariant transformer, small 1 layer, hidden 8, 2 heads



$$c_{ij} = Wh_i + Uh_j + V\|x_i - x_j\|^2 \qquad h \leftarrow h + \mathrm{ReLU}(W[h, \textstyle\sum_j \bar{\alpha}_{ij} c_{ij}])$$

$$v \leftarrow v + \textstyle\sum_j \tanh(u^\top c_{ij})\, \bar{\alpha}_{ij}\, (x_i - x_j) \qquad \bar{\alpha}_{ij}\text{: attention weight, averaged over heads}$$

Position error validation MSE over training, big and small networks



Numbers beside each name: validation MSE at epoch 100, big / small.

Probe scores: EGNN

Fact	Embed	Block 1	Block 2	Block 3	Block 4
Charge	1.000	1.000	1.000	1.000	1.000
Distance		0.999	0.982	0.957	0.800
Charge product		0.997	0.362	0.432	0.132
Force magnitude		0.951	0.561	0.468	0.345
Potential energy		0.982	0.897	0.820	0.734
Force direction		0.017	−0.004	0.004	0.004

Validation R^2 of a linear probe. Force direction is a mean cosine, not an R^2.

Probe scores: Attention EGNN

Fact	Embed	Block 1	Block 2	Block 3	Block 4
Charge	1.000	1.000	1.000	1.000	1.000
Distance		0.397	0.741	0.759	0.502
Charge product		0.273	0.480	0.509	0.257
Force magnitude		0.813	0.645	0.706	0.788
Potential energy		0.588	0.761	0.700	0.742
Force direction		0.021	0.026	0.045	0.084

Validation R^2 of a linear probe. Force direction is a mean cosine, not an R^2.

Probe scores: Non-equivariant GNN

Fact	Embed	Block 1	Block 2	Block 3	Block 4
Charge	1.000	1.000	1.000	0.989	0.981
Distance		0.993	0.968	0.944	0.695
Charge product		0.915	0.233	0.091	0.003
Force magnitude		0.860	0.400	0.393	0.240
Potential energy		0.896	0.732	0.632	−2.344
Force direction		0.018	−0.004	0.004	−0.004

Validation R^2 of a linear probe. Force direction is a mean cosine, not an R^2.

Probe scores: MLP

Fact	Block 1	Block 2	Block 3	Block 4
Charge	0.909	0.733	0.520	0.345
Distance	0.191	0.106	0.038	−0.013
Charge product	−0.007	−0.033	−0.051	−0.062
Force magnitude	−0.024	−0.048	−0.061	−0.075
Potential energy	−0.026	−0.013	−0.065	−0.178

Validation R^2 of a linear probe.

Probe scores: Hamiltonian network

Fact	Embed	Block 1	Block 2
Charge	1.000		
Distance		1.000	1.000
Charge product		0.012	1.000
Force magnitude		0.650	0.938
Potential energy		0.056	0.887
Net force		0.085	

Validation R^2 of a linear probe. Net force is a mean cosine, not an R^2.

Probe scores: Equivariant transformer

Fact	Embed	Block 1	Block 2	Block 3	Block 4
Charge	1.000	1.000	1.000	1.000	1.000
Distance		0.937	0.938	0.941	0.943
Charge product		0.000	0.005	0.020	0.040
Force magnitude		0.264	0.267	0.274	0.304
Potential energy		0.000	0.034	0.092	0.360
Force direction		0.129	0.096	0.076	0.041

Validation R^2 of a linear probe. Force direction is a mean cosine, not an R^2.

Deletions

Network	Removed	Before	After
Non-equivariant GNN	Force magnitude, block 1	0.079	0.458
Hamiltonian network	Distance, block 2	0.126	0.323
Equivariant transformer	Distance, block 1	0.051	0.216
Non-equivariant GNN	Distance, block 1	0.079	0.114
Non-equivariant GNN	Charge product, block 1	0.079	0.102
Attention EGNN	Charge, block 2	0.014	0.028
EGNN	Charge product, block 1	0.090	0.103
Hamiltonian network	Distance, block 1	0.126	0.136
Non-equivariant GNN	Potential energy, block 1	0.079	0.086
Attention EGNN	Charge, embed	0.014	0.019
MLP	Charge, block 1	0.222	0.211
Hamiltonian network	Charge, embed	0.126	0.120

Validation MSE before and after the probe direction is removed.

Small networks

Network	Error	Charge	Distance	$q_i q_j$	$\|F\|$	Potential	Force
EGNN	0.129	1.000	0.222	0.000	0.026	−0.002	−0.225
Attention EGNN	0.125	1.000	−1.230	0.276	0.245	−4.651	0.387
Non-equivariant GNN	0.131	1.000	0.011	0.002	0.008	−0.009	0.002
Equivariant transformer	0.085	1.000	0.937	0.000	0.264	0.000	0.788
Hamiltonian network	0.125	1.000	0.907	0.990	0.478	0.754	0.214
MLP	0.140	0.038	0.001	−0.003	−0.007	−0.005	

Error is validation MSE at epoch 100. Force is a mean cosine. The other columns are probe R^2.

Small-network deletions

Network	Removed	Before	After
Equivariant transformer	Charge, embed	0.085	0.298
Equivariant transformer	Distance, block 1	0.085	0.245
Attention EGNN	Charge, embed	0.125	0.137

Validation MSE before and after the probe direction is removed.

Assignment 2

Assignment 2 run	Network here	Assignment 2	Here
Attention	Attention EGNN	0.0103	0.014
Full EGNN	EGNN	0.0945	0.090
No equivariance	Non-equivariant GNN	0.0579	0.079
No residual		0.0896	
No velocity		0.0983	
No distance		0.1279	
No tanh		11.46	

Validation MSE.